Appendix 2 *for:* Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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DEB model parameters and example simulations

1 DEB core parameter analysis across individuals

We estimated Dynamic Energy Budget (DEB) model parameters for 37 *I. alpestris* individuals using the standard DEB model (std) (Kooijman, 2010). The standard DEB model is governed by 12 primary parameters that jointly determine the rates of assimilation, mobilisation, maintenance, growth, and reproduction (Marques et al., 2018). For the population-average parameterisation, seven of the 12 primary parameters were freed for estimation; the remaining five were held at their literature values. At the individual level, four parameters were allowed to vary: the zoom factors z (female) and z_m (male), the energy conductance $\dot{v}$, and the volume-specific somatic maintenance rate $[\dot{p}_M]$. All remaining parameters were held at their population-average values.

The zoom factor z scales maximum body size ($L_m = z \cdot L_{\text{ref}}$, with $L_{\text{ref}} = 1$ cm) and, through the physical covariation rules (Kooijman, 2010), simultaneously determines the maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$ and the maturity thresholds at birth (E_H^b) and puberty (E_H^p). Because z (or z_m) and $[\dot{p}_M]$ vary among individuals while κ is fixed, $\{\dot{p}_{Am}\}$ is not a free parameter but is derived from the individual estimates of z and $[\dot{p}_M]$:

$$\{\dot{p}_{Am}\} = \frac{z \cdot L_{\text{ref}} \cdot [\dot{p}_M]}{\kappa} \qquad \text{and} \qquad \{\dot{p}_{Am,m}\} = \frac{z_m \cdot L_{\text{ref}} \cdot [\dot{p}_M]}{\kappa} \tag{1}$$

The complete set of parameter values is presented in **Table S2.1**.

Table S2.1: DEB parameter values for *I. alpestris*: population-average estimates and individual ranges (min–max) across the 37 individuals.

Parameter	Symbol	Units	Population average	Individuals	
				Min	Max
Primary parameters					
max spec searching rate	$\{\dot{F}_m\}$	$\mathrm{l\,d^{-1}\,cm^{-2}}$	6.5	–	–
digestion efficiency of food to reserve	κ_X	–	0.8	–	–
max surface-specific assimilation rate	$\{\dot{p}_{Am}\}$	$\mathrm{J\,d^{-1}\,cm^{-2}}$	265.7	217.6	303.3
energy conductance	$\dot{v}$	$\mathrm{cm\,d^{-1}}$	0.01914	0.01716	0.02134
allocation fraction to soma	κ	–	0.5914	–	–
reproduction efficiency	κ_R	–	0.95	–	–
vol-spec somatic maint	$[\dot{p}_M]$	$\mathrm{J\,d^{-1}\,cm^{-3}}$	138.4	107.4	162.8
surf-spec somatic maint	$\{\dot{p}_T\}$	$\mathrm{J\,d^{-1}\,cm^{-2}}$	0	–	–
maturity maint rate coefficient	$\dot{k}_J$	$\mathrm{d^{-1}}$	0.002	–	–
spec cost for structure	$[E_G]$	$\mathrm{J\,cm^{-3}}$	7302	–	–
maturity at birth	E_H^b	J	6.126	–	–
maturity at puberty for females	E_H^p	J	1.366×10^4	–	–
Extended parameters					
zoom factor	z	–	1.135	1.102	1.198
zoom factor for males	z_m	–	1.046	0.9978	1.118
max surface-specific assimilation rate	$\{\dot{p}_{Am,m}\}$	$\mathrm{J\,d^{-1}\,cm^{-2}}$	244.8	191.9	307.9
maturity at puberty for males	$E_{H,m}^p$	J	7474	–	–
maturity at metam	E_H^j	J	334.2	–	–
shape coefficient	δ_M	–	0.1878	–	–
Weibull aging acceleration	$\ddot{h}_a$	$\mathrm{d^{-2}}$	6.475×10^{-9}	–	–

1.1 Parameter variability

The coefficient of variation (CV) quantifies relative variability across individuals. **Figure S2.1** shows the CV values sorted in descending order. Among the four free parameters, $[\dot{p}_M]$ exhibited the largest relative variation across individuals, followed by $\dot{v}$, while z was the most conservative. The mean individual values deviated from the population-average estimate by 0.9% for z, -0.9% for z_m, 1% for $\dot{v}$, and -0.8% for $[\dot{p}_M]$.

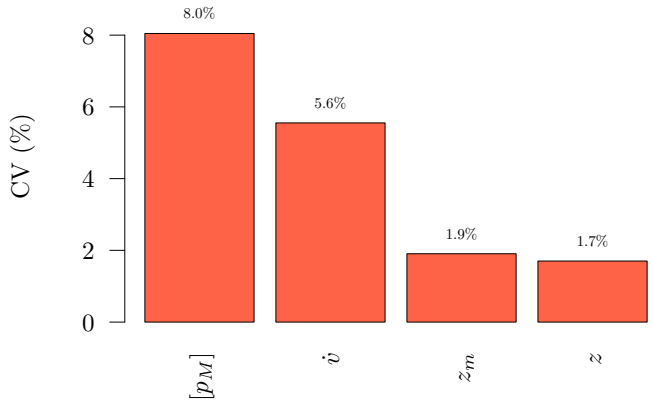


Figure S2.1: Coefficient of variation for each varying parameter. $[\dot{p}_M]$ is the most variable parameter across individuals.

Figure S2.2 shows the distribution of each parameter with the population-average value marked for reference. Most parameters were broadly distributed around the

population-average estimate, indicating that the individual-level data span a wider range of physiological strategies than the single population-average value suggests.

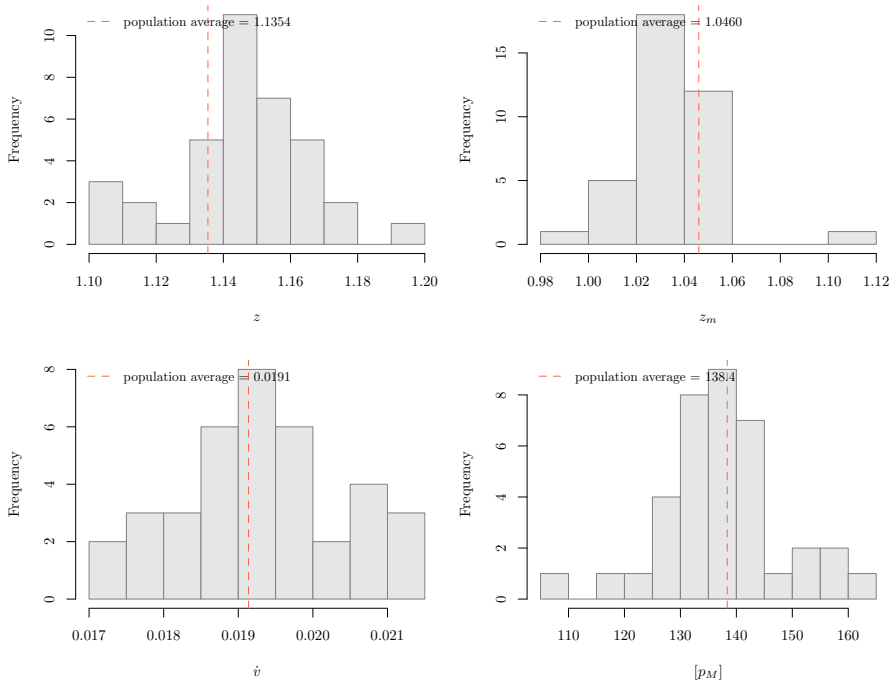


Figure S2.2: Distribution of each varying parameter. The dashed line marks the population-average estimate.

2 Fit of DEB models to the data

In **Fig. S2.3** and **Fig. S2.4** we show the fit of the model with the final set of estimated parameters for the population-average parameterisation (for each sex in different colours) and for four selected females and males.

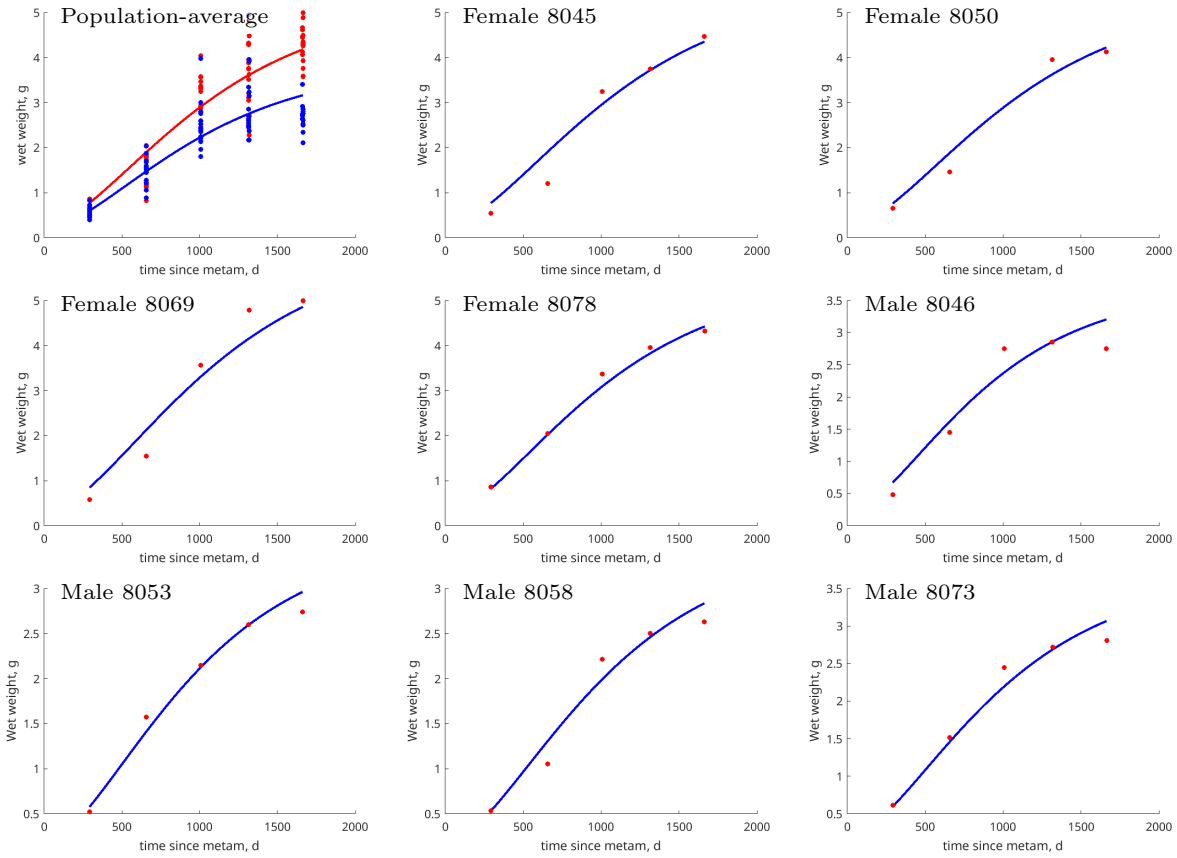


Figure S2.3: Model predictions of growth for the population-average parameterisation (*blue*: males; *red*: females) and eight selected individuals.

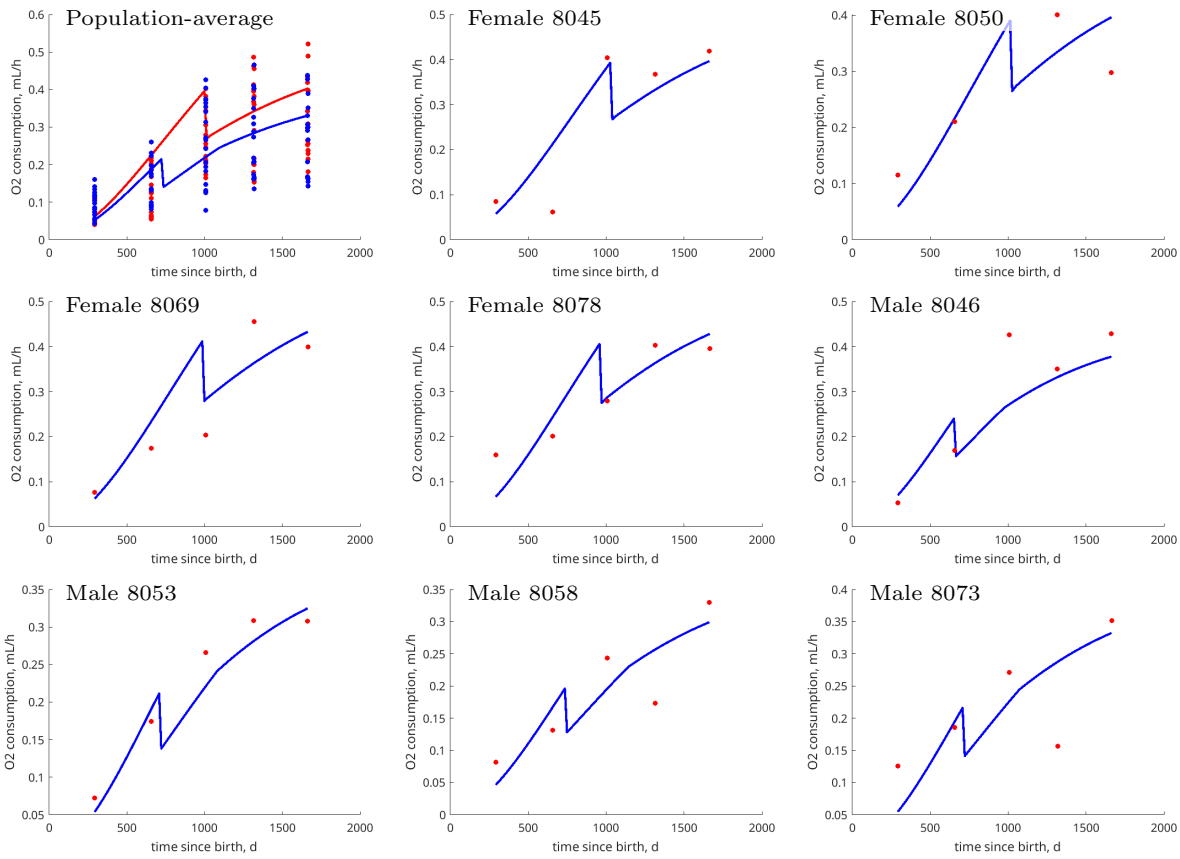


Figure S2.4: Model predictions for metabolic rates for the population-average parameterisation (*blue*: males; *red*: females) and eight selected individuals.

3 Growth trajectories

Individual 20-year DEB growth simulations at the **(a)** Czech reference site (49.2247° N, 16.0579° E; 600 m a.s.l.) and **(b)** at a comparatively warmer site in Northern Spain (inside the known distribution of the species). A vertical dashed line marks when individuals reached puberty (first hour where maturity E_H exceeds the puberty threshold E_H^p). We present example simulations for females first (with the population-average parameters, and for four selected individuals) and then for males (population-average parameters first and for individual-level parameters after). Note that the selected individuals correspond to the individuals shown in **Figs.** S2.3 and S2.4.

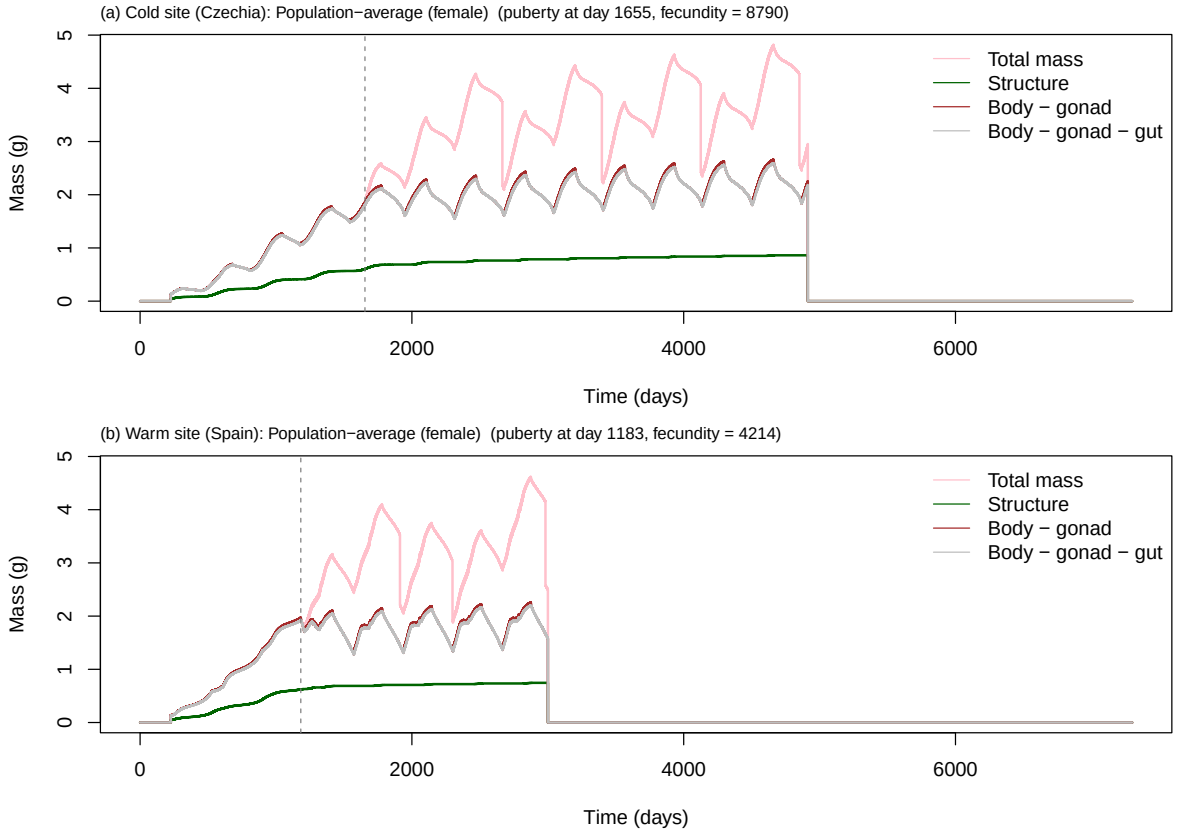


Figure S2.5: Growth trajectory for the population-average parameterisation (female).

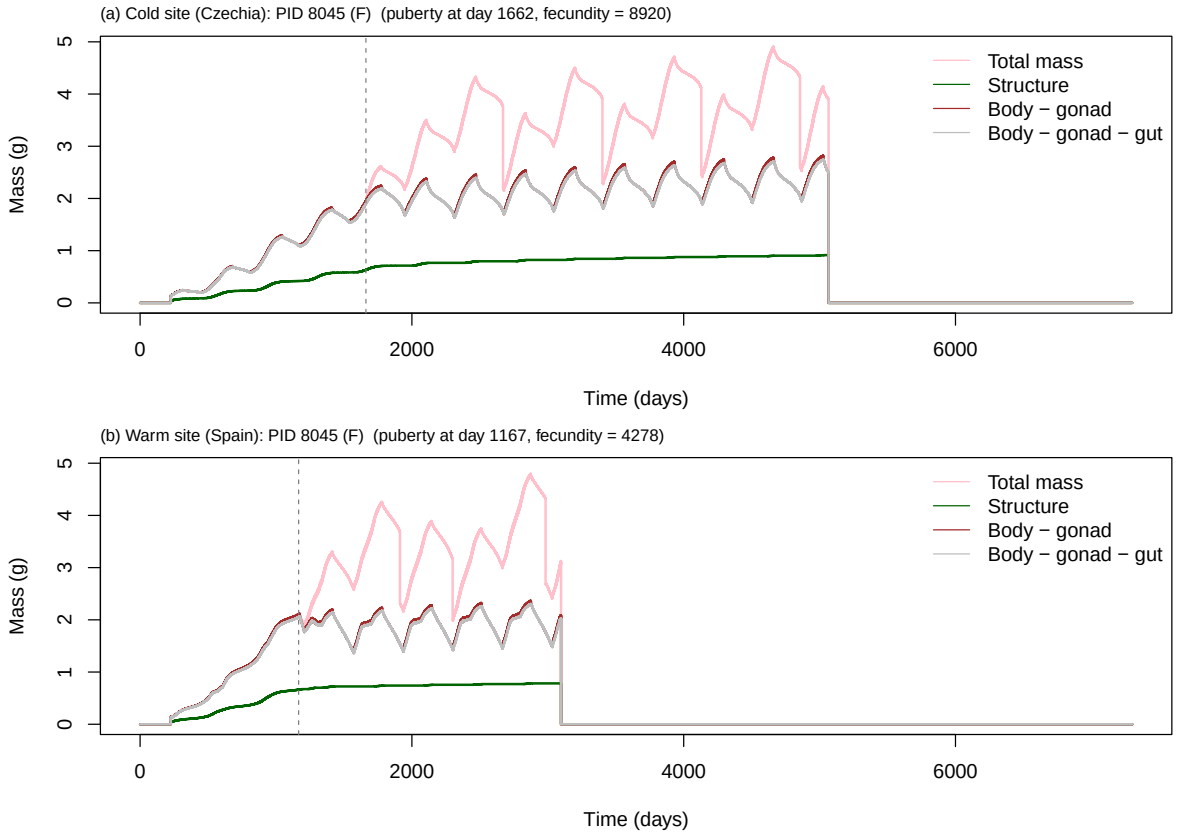


Figure S2.6: Growth trajectory for PID 8045 (female).

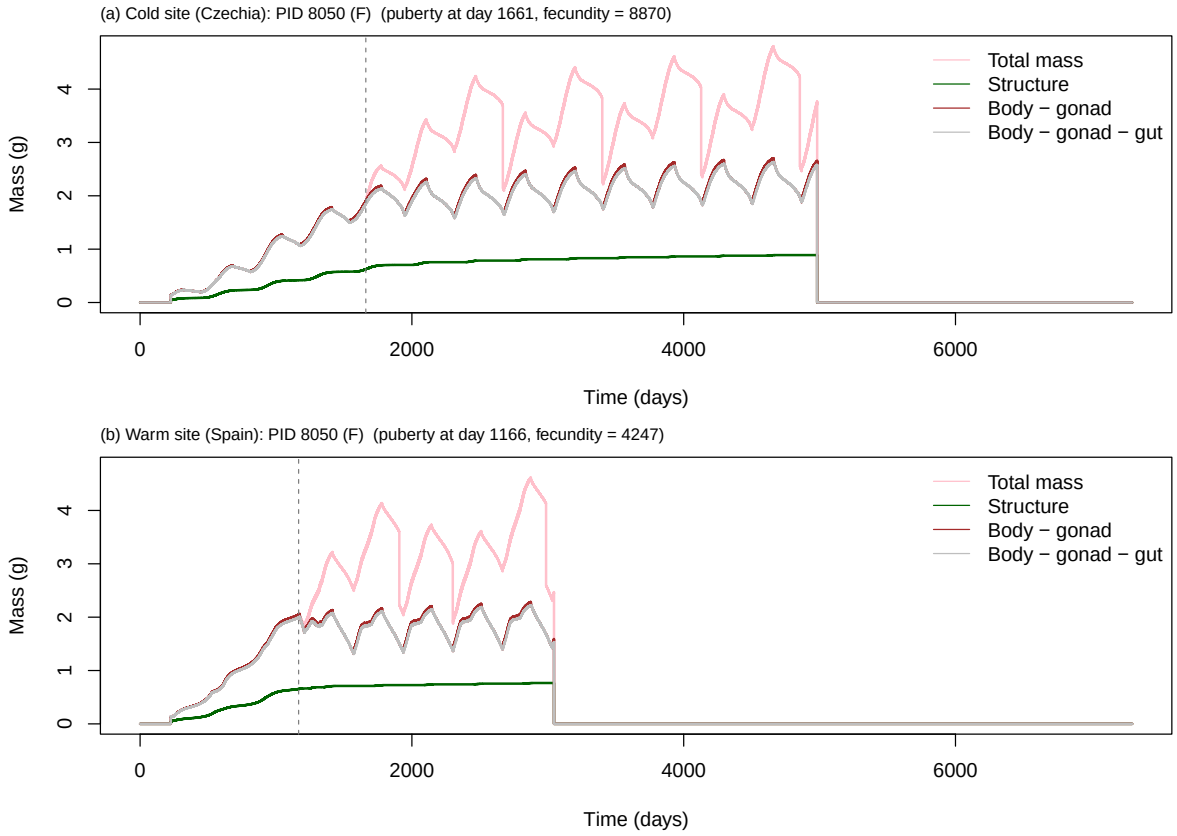


Figure S2.7: Growth trajectory for PID 8050 (female).

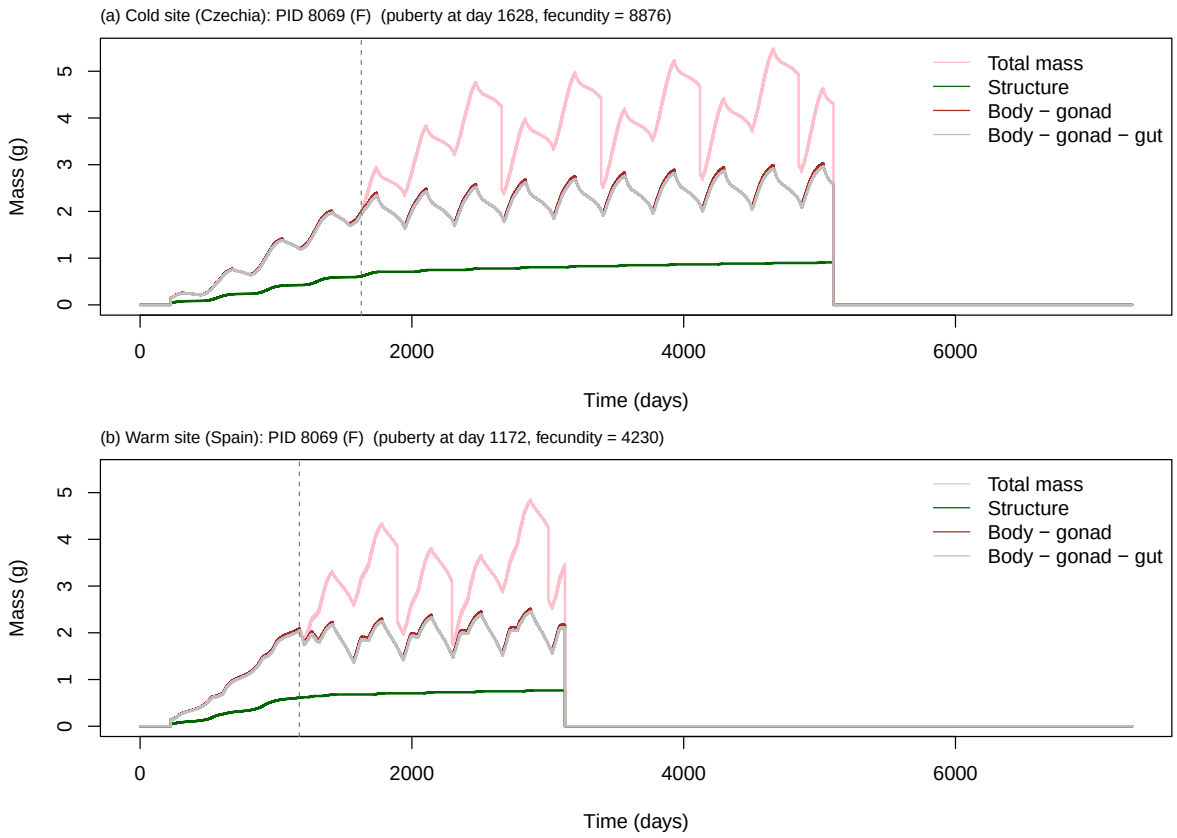


Figure S2.8: Growth trajectory for PID 8069 (female).

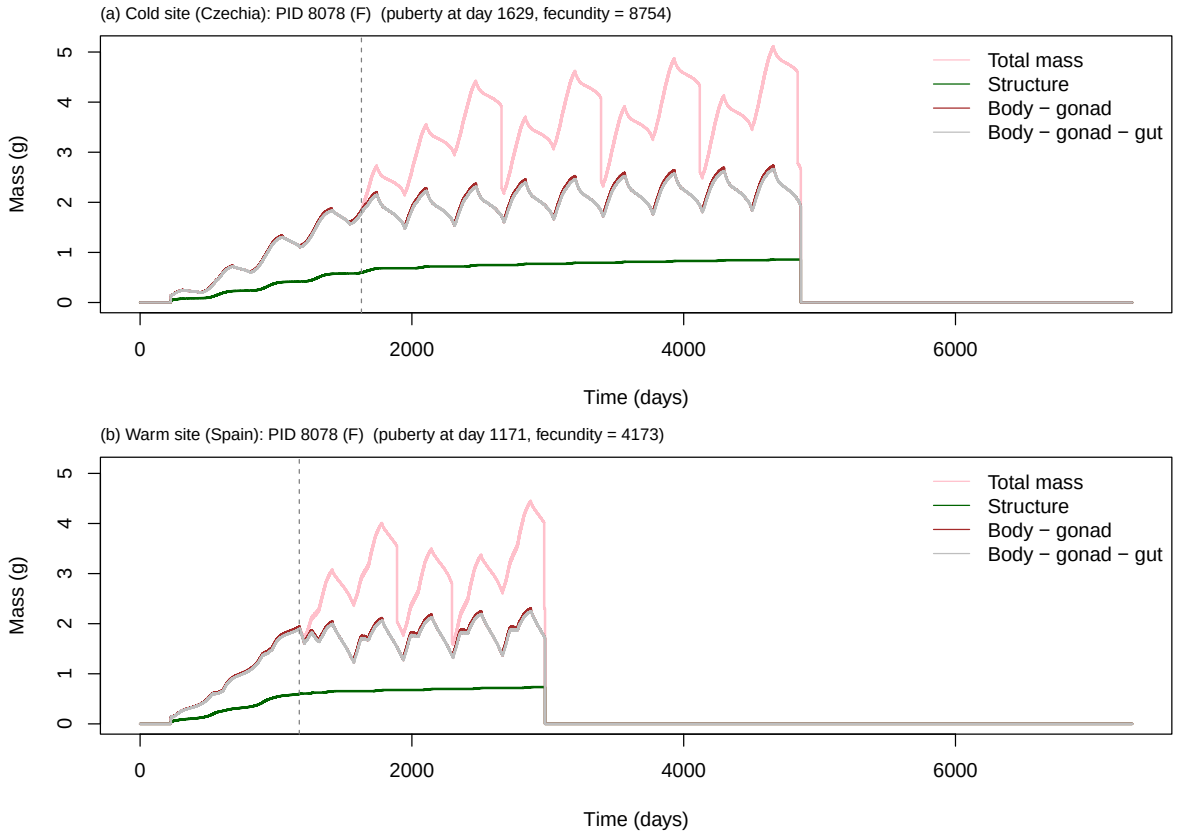


Figure S2.9: Growth trajectory for PID 8078 (female).

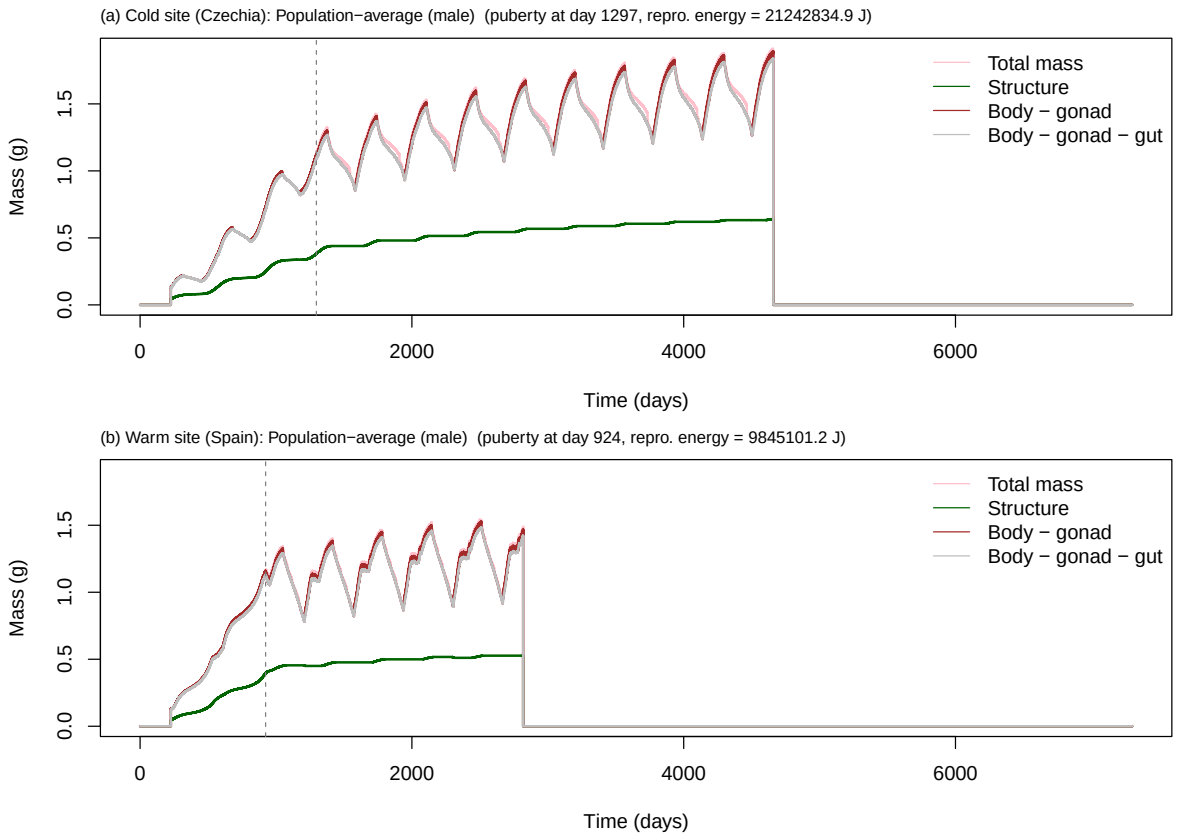


Figure S2.10: Growth trajectory for the population-average parameterisation (male).

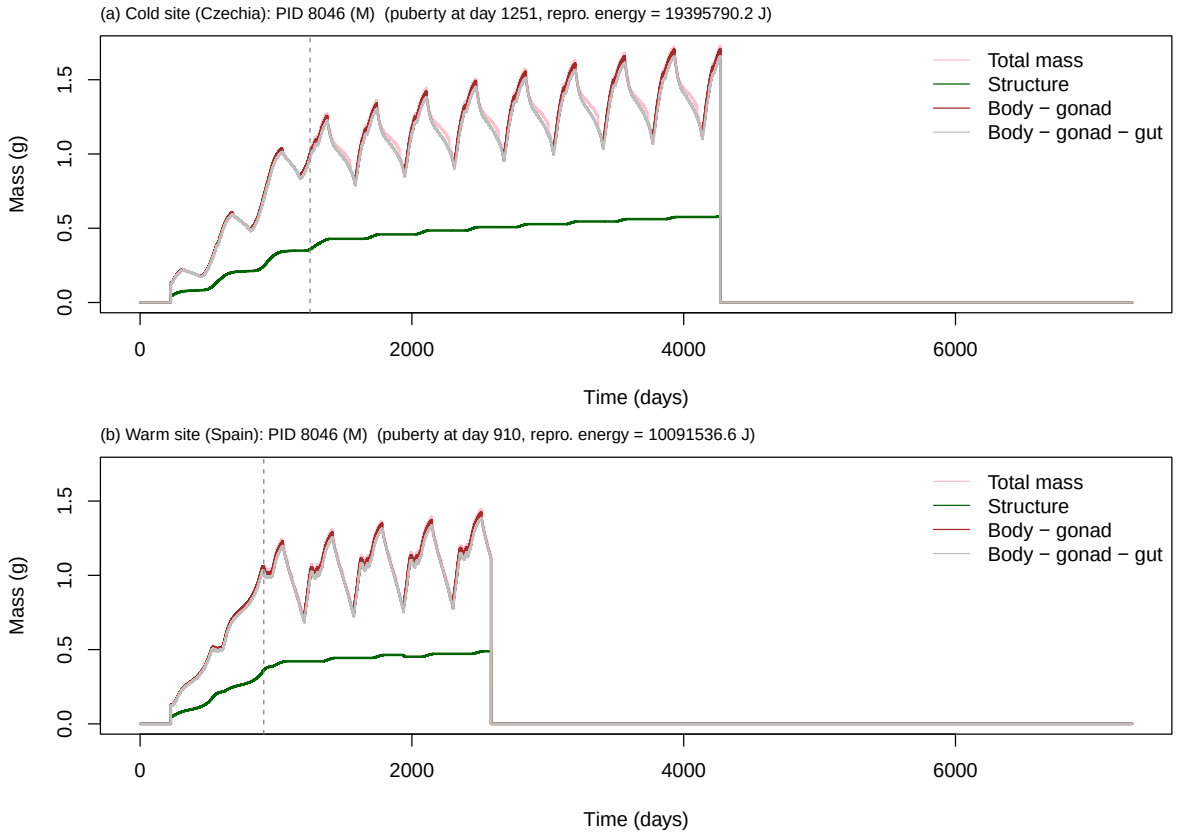


Figure S2.11: Growth trajectory for PID 8046 (male).

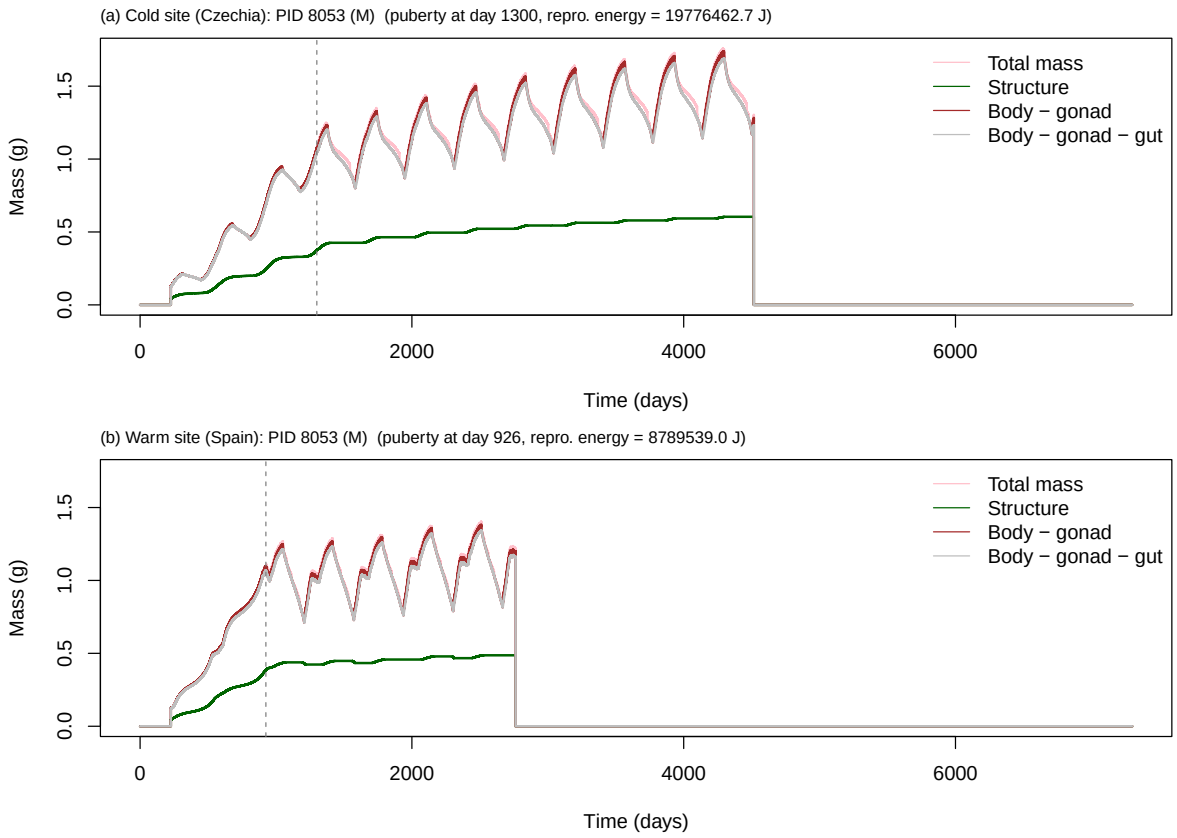


Figure S2.12: Growth trajectory for PID 8053 (male).

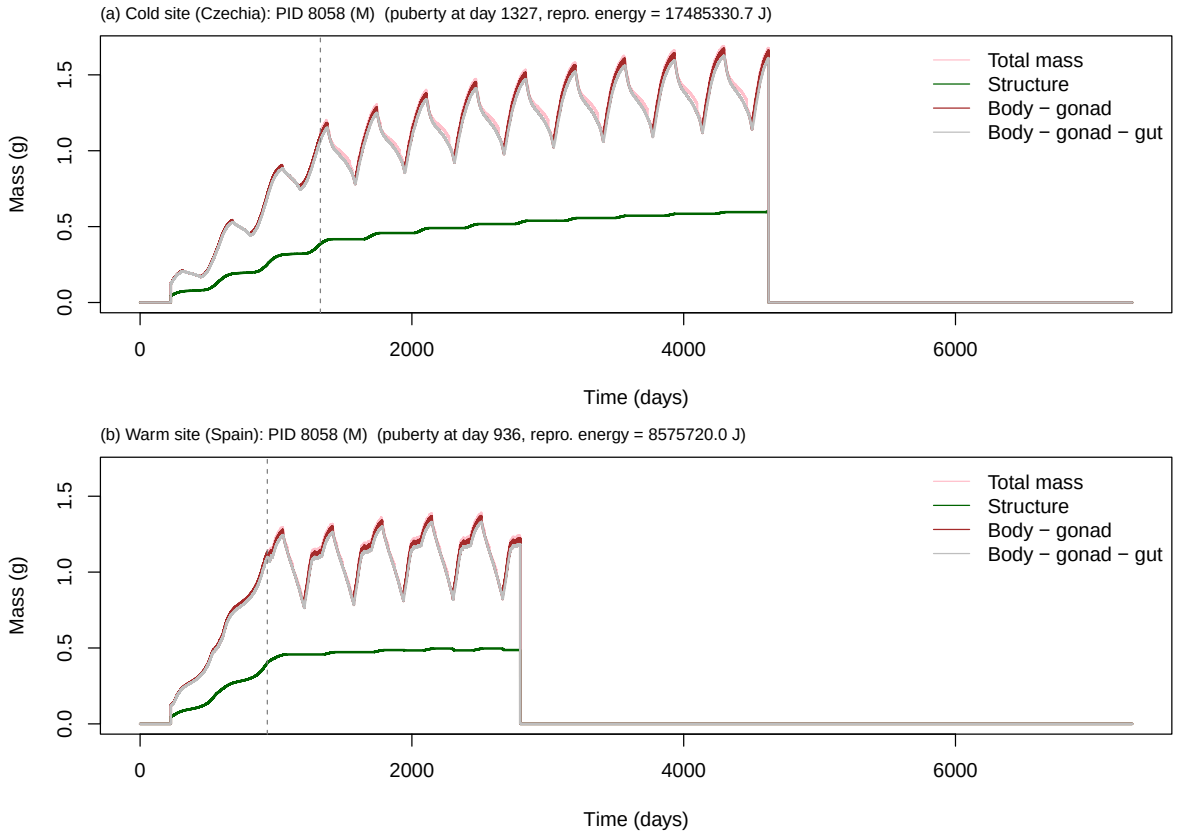


Figure S2.13: Growth trajectory for PID 8058 (male).

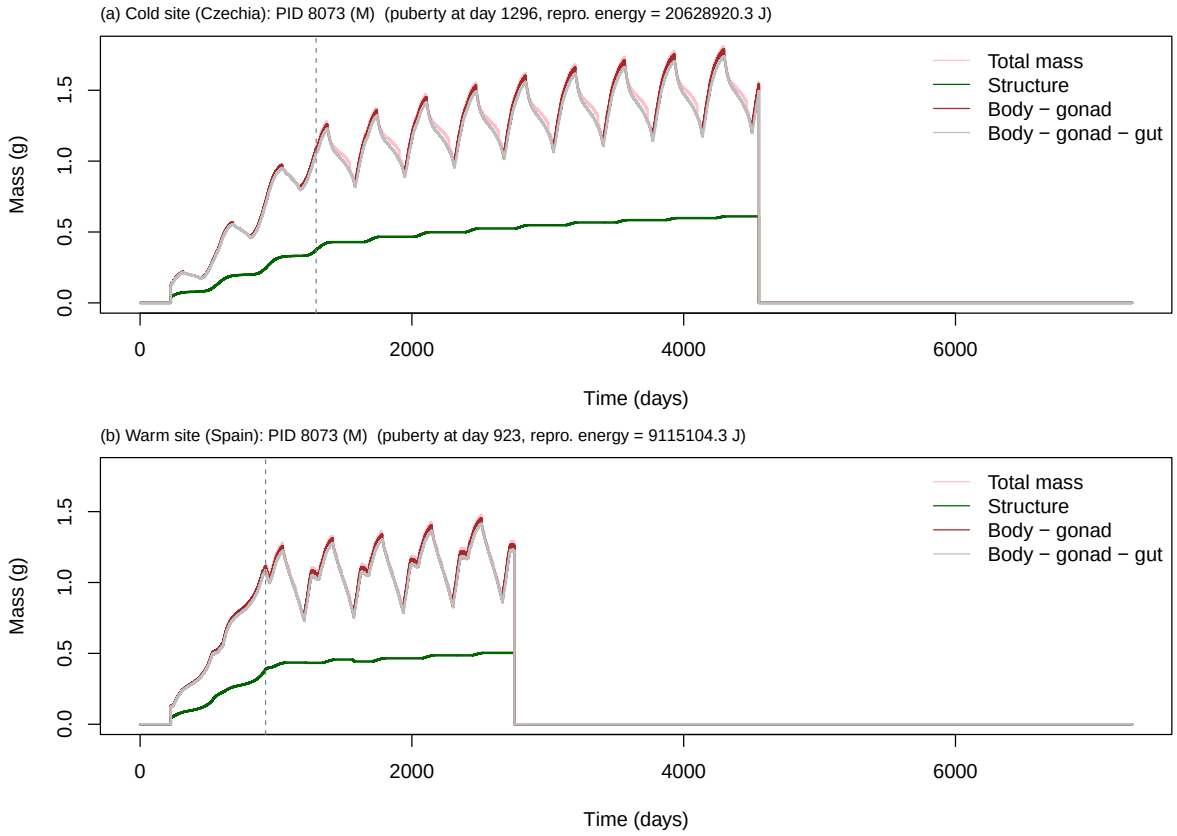


Figure S2.14: Growth trajectory for PID 8073 (male).

4 References

Kooijman, S. A. L. M. (2010). *Dynamic energy budget theory for metabolic organisation* (3rd ed.). Cambridge university press.

Marques, G. M., Augustine, S., Lika, K., Pecquerie, L., Domingos, T., & Kooijman, S. A. L. M. (2018). The AmP project: Comparing species on the basis of dynamic energy budget parameters (D. A. Beard, Ed.). *PLOS Computational Biology*, *14*(5), e1006100. https://doi.org/10.1371/journal.pcbi.1006100